

# Evaluating Functions from Rules, Tables, and Graphs

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*Substituting into rules, reading tables, reading graphs, and comparing outputs*

**Directions**

- $f(3)$  means “the output of  $f$  when the input is 3”. It is never  $f$  multiplied by 3.
- Show the substitution you made, or point to the table row or plotted point you read.
- In Part B every problem has its own graph. Read values only from the marked points on that graph.

**Part A — rules and tables**

1.  $f(x) = 4x - 3$ . Find  $f(5)$ .

Answer: \_\_\_\_\_

2.  $g(x) = -2x + 9$ . Find  $g(-3)$ .

Answer: \_\_\_\_\_

3.  $h(x) = x^2 - 4x$ . Find  $h(-2)$ .

Answer: \_\_\_\_\_

4. The table gives the function  $p$ . Find  $p(3)$ .

|        |   |    |    |    |    |
|--------|---|----|----|----|----|
| $x$    | 0 | 1  | 2  | 3  | 4  |
| $p(x)$ | 7 | 11 | 15 | 19 | 23 |

Answer: \_\_\_\_\_

5. A theatre charges the same price for every ticket. The table gives  $c(n)$ , the cost in dollars of  $n$  tickets. Find  $c(4)$ .

|        |    |    |    |    |    |
|--------|----|----|----|----|----|
| $n$    | 1  | 2  | 3  | 4  | 5  |
| $c(n)$ | 12 | 24 | 36 | 48 | 60 |

Answer: \_\_\_\_\_

6. Use  $f(x) = 4x - 3$  from problem 1 and the table for  $p$  from problem 4. Fill in the two outputs, then write  $<$ ,  $>$ , or  $=$  in the last blank.

$$f(3) = \underline{\hspace{2cm}} \quad p(3) = \underline{\hspace{2cm}} \quad f(3) \underline{\hspace{1cm}} p(3)$$

7.  $f(x) = 3x + 2$ . Find  $f(-4) = \underline{\hspace{2cm}}$

Now explain in one or two sentences why  $f(-4)$  is not the same as  $-f(4)$ .

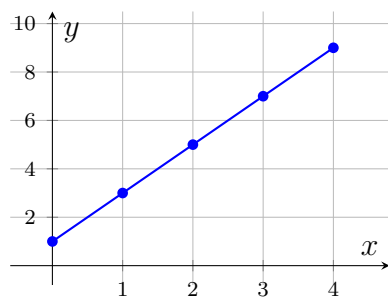
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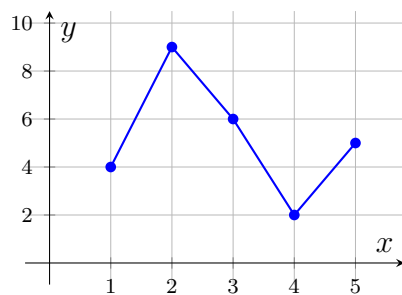
## Part B — graphs

8. The graph of  $q$  is shown. Find  $q(3)$ .



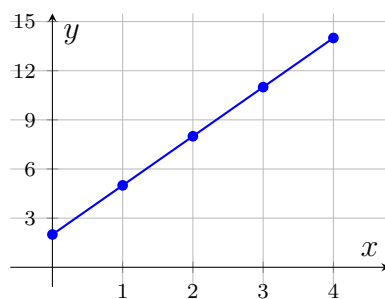
Answer: \_\_\_\_\_

9. The graph of  $r$  is shown. For which input is the output of  $r$  greatest?



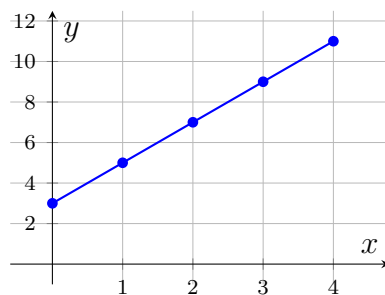
Answer: \_\_\_\_\_

10. The graph of  $s$  is shown. Fill in all three blanks.



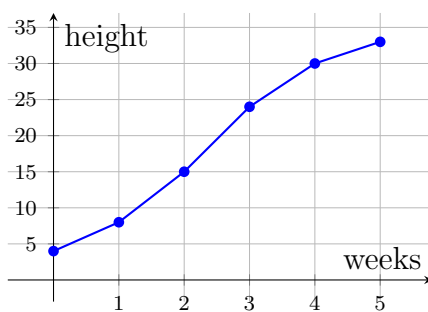
$$s(1) = \underline{\hspace{2cm}} \quad s(4) = \underline{\hspace{2cm}} \quad s(4) - s(1) = \underline{\hspace{2cm}}$$

11. The graph shows the function  $m$ . A second function is given by the rule  $n(x) = 4x - 1$ . Fill in the two outputs, then write  $<$ ,  $>$ , or  $=$  in the last blank.



$$m(2) = \underline{\hspace{2cm}} \quad n(2) = \underline{\hspace{2cm}} \quad m(2) \underline{\hspace{1cm}} n(2)$$

12. A sunflower seedling was measured once a week. The graph shows  $w$ , its height in centimetres, as a function of the number of weeks since the first measurement. Find  $w(3)$ , then explain what that value tells you about the seedling.



Answer: \_\_\_\_\_ cm

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